

Modeling and Simulation

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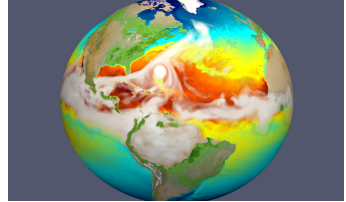
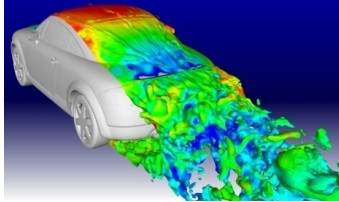
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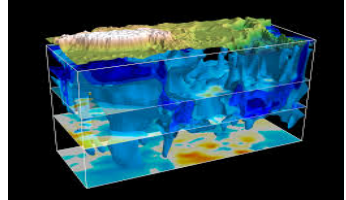
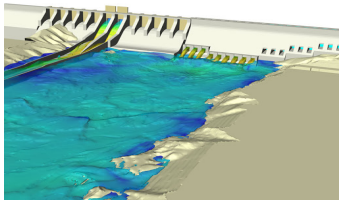
Introduction and General Concepts

Motivational Examples: Modeling and Simulating Real Processes

1. Physics, Astrophysics, Climatology, ...



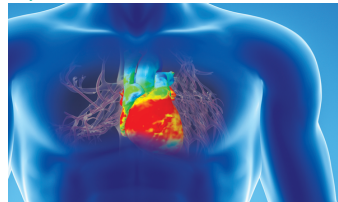
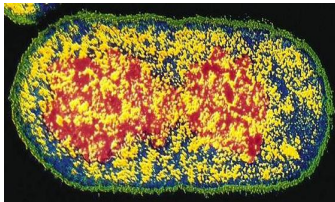
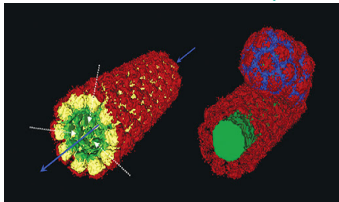
2. Environmental Systems (Rivers, Volcano, Earthquakes, ...)



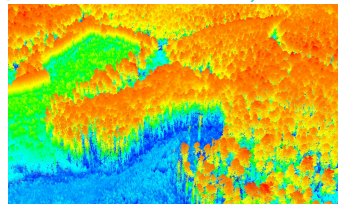
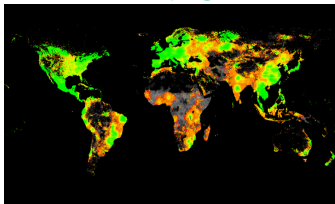
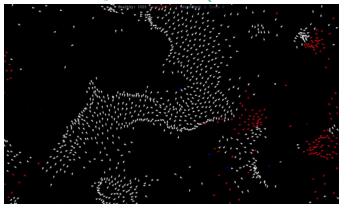
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Motivational Examples: Modeling and Simulating Real Processes

3. Biology Systems (Tissues Growth, Cells, Organs, ...)



4. Ecosystems (Swarm behaviours, Propagation of Epidemic, Forests,...)



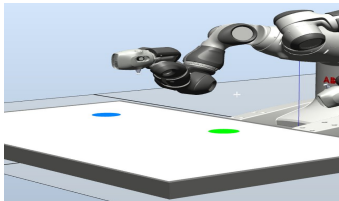
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Motivational Examples: Modeling and Simulating Real Processes

5. Social Systems (Pedestrian Motion, Traffic, ...)



6. Robotics and Autonomous Self Driving cars

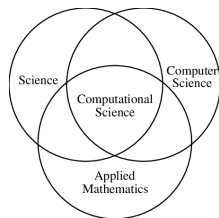


7. And others,...

Introduction and General Concepts

Modeling and Simulation as a methodology in Computational Science

1. Computational Science, Modeling and Simulation, and Computational Scientist



- **Computational Science (CS)** is a multi-disciplinary emerging field that puts the computer at the center of the methodology.
- **Modeling and Simulation (M&S)** is typical example of this methodology.
- **M&S:** Discipline of understanding and evaluating the interaction of parts of a real or theoretical system by:
 - Designing its representation (model)
 - Executing (running) the model including the time and space dimension (simulation).
- So **CS** is at the intersection between several field like computer science, like mathematics, like physics,
- A **Computational Scientist** needs to be a bit of all.

Introduction and General Concepts

Advantages/Disadvantages of doing Modeling and Simulation

1. Advantages

- **Easy to understand** — Allows to understand how the system really operates without working on real-time systems.
- **Easy to test** — Allows to make changes into the system and their effect on the output without working on real-time systems.
- **Easy to upgrade** — Allows to determine the system requirements by applying different configurations.
- **Easy to diagnose problems** — Allows to understand all the interactions and analyze their effect. Additionally, new policies, operations, and procedures can be explored without affecting the real system.

Introduction and General Concepts

Advantages/Disadvantages of doing Modeling and Simulation

2. Disadvantages

- Designing a model is an art which requires domain knowledge, training and experience.
- Operations are performed on the system using random number, hence difficult to predict the result.
- Simulation requires manpower and it is a time-consuming process.
- Simulation results are difficult to translate. It requires experts to understand.
- Simulation process is expensive.

Introduction and General Concepts

Key Modeling and Simulation Concepts

1. What is a model?



- This is not an apple. It is just its graphical representation.
- It looks very much like an apple (You cannot eat it or cut it into pieces).
- You can just look at it. It's just the representation of an apple.
- So this image is perfect to model an apple. But if you would like someone to test it, of course this is not sufficient.
- So a **model** is just a **representation of reality** , or at least a **representation of a part of reality**, but not the full system.

Introduction and General Concepts

Key Modeling and Simulation Concepts

1. What is a model?



Many Possible formal definition

- Simplified abstraction of a reality, allowing us to better describe and understand it.
- An abstraction in which only the essential ingredients are retained, according to the question we ask for the system.
- It is the representation of a phenomena in a mathematical or computer-based language.
- A physical, mathematical, or otherwise logical representation of a system, entity, phenomenon, or process.

2. Why a model?

Several reasons:

- Describe, classify, but mostly,
- Understand how things work.
- Predict how a system will involve
- Control a Phenomena

Introduction and General Concepts

Key Modeling and Simulation Concepts

3. What is the difference between modeling and simulation?

Modelling:

- The process of representing a model which includes its construction and working.
- creating a model which represents a system including their properties. It is just an act of building a model.

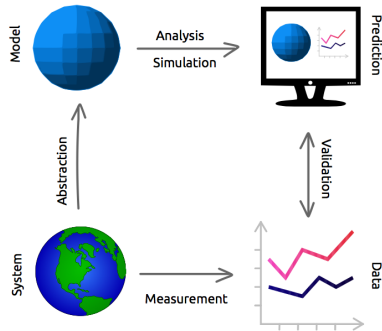
Simulation:

- The operation of a model in terms of time or space.
- The process of using a model to study the performance of a system. It is an act of using a model for simulation.

Introduction and General Concepts

Key Modeling and Simulation Concepts

4. From System to Model to Simulation

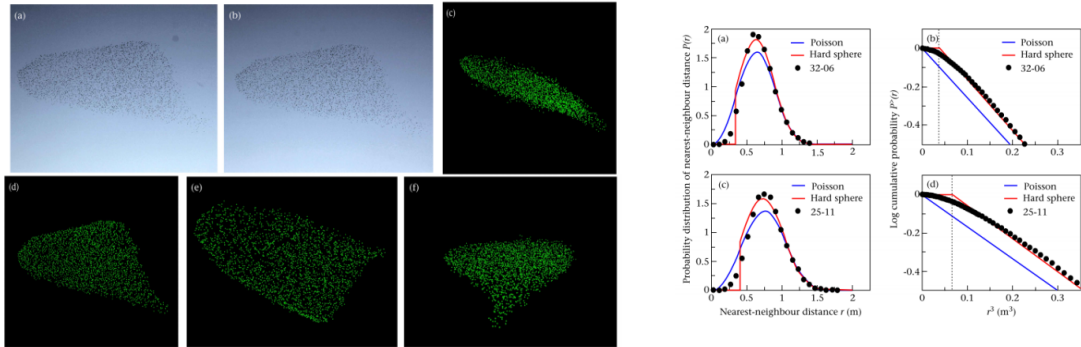


- **System:** something in the real world we are interested in.
- **Abstraction:** deciding which details can be left out from the system.
- **Model:** a description of the system that includes only the features we think are essential.
- The model can be studied using mathematical **analysis** or implemented as computer program, which can run **simulations**.
- The result of analysis and simulation might be a **prediction** about what the system will do.
- predictions can be **validated** by comparing measurements from the real world with the results from analysis and simulation.

Introduction and General Concepts

Key Modeling and Simulation Concepts

5. An illustration Example: Starling flock behaviour



Huemiller, Erik, and Nigel Goldenfeld. "The Emergence of Polarization in Flocks of Starlings."

Introduction and General Concepts

Key Modeling and Simulation Concepts

Simulation Model Components - Definitions

- **Entities:** An entity represents a physical (or logical) object in your system that must be explicitly captured in the model in order to be able to describe the overall operation of the system.
- **Attributes:** An entity is characterized using attributes, which are local variables defined inside the entity.
- **State Variables:** The state of the system under study is represented by a set of state variables. A state variable is used to track a property of a static entity over time.
- **Events:** An event represents the occurrence of something interesting inside the system. It is a stimulus that causes the system to change its state.
- **Activities:** An activity is an action which is performed by the system for a finite duration of time.

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Key Modeling and Simulation Concepts

Simulation Model Components - Illustration Example

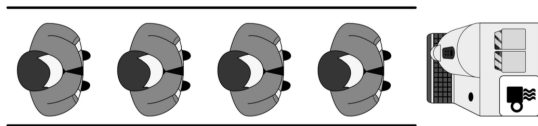


Figure 1: Coffee queuing system.

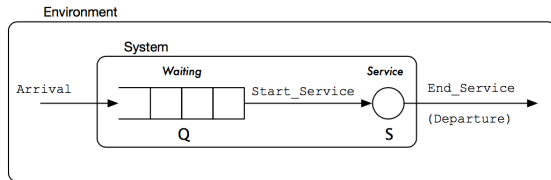


Figure 2: Coffee queuing model.

Element	Details
Entity	Queue, Server, Person
State Variables	Q = Number of Persons in Queue $Q \in \{0, 1, 2, \dots\}$ S = Status of Server $S \in \{\text{Free, Busy}\}$
Events	Arrival, Start_Service, End_Service (or Departure)
Activities	Generation, Waiting, Service, Delay

Figure 3: Details of the Coffee queuing model.

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Simulation Models Classification

Simulation models can be classified along three different dimensions:

- **Static vs. Dynamic Simulation Models** - A static simulation model is a representation of a system at a particular time, or one that may be used to represent a system in which time simply plays no role (e.g. Monte Carlo models). On the other hand, a dynamic simulation model represents a system as it evolves over time, such as a conveyor system in a factory.
- **Deterministic vs. Stochastic Simulation Models** - If a simulation model does not contain any probabilistic (i.e., random) components, it is called deterministic (e.g. differential equations describing a chemical reaction might be such a model. Many systems, however, must be modeled as having at least some random input components, and these give rise to stochastic simulation models (e.g. queueing and inventory systems).
- **Continuous vs. Discrete Simulation Models** - Continuous system is affected by the state variable, which changes continuously as a function with time (e.g. car moving on a road represents a continuous system). Discrete system is affected by the state variable changes at a discrete point of time.

Introduction and General Concepts

Modeling Space and Time

Time and Space

- Systems evolve in time and the state of the system may change in place (space and time dependent). e.g:
 - The atmospheric pressure or temperature are different from one place to another and they change during the day and the next day.
 - A car on a road changes position as time goes on.
- Sometime, we don't need to consider space in a model. e.g.:
 - Interested only in the time evolution of global quantity like how many individual of a given species I have in an area (counting the number of individuals in a population).
 - Interested only in the time evolution of the total amount of CO₂ in the atmosphere
- Some models can remove the temporal dimension if time doesn't evolve, e.g:
 - The temperature in the room, you maybe interested whether it's cold near the window or warm in the middle

Introduction and General Concepts

Modeling Space and Time

Time Dimension

Temporal dimension can be captured in a model using:

1. **Continuous Time** - Time is a continuous variable (as Physics suggests). Only Differential Equations-based mathematical models can deal with the continuous time.
2. **Discrete Time** - Once comes to the computer, we have to change this continuous time by discretizing it:
 - The duration of the process is broken into small time steps Δt
 - Then describe the state of the system at each time step: $t_0 = 0, t_1 = \Delta t, \dots, t_n = n\Delta t, \dots$
3. **Discrete Event Time** - Alternatively, we can only focus on the interesting moments of a process (Discrete Events Simulation (DES)).
 - In a queue in a post office, we don't really care to describe the evolution of the system when nothing happen (Just capture the moment when your customer comes? Or when one done).
 - The time t at which an event occurs can be any real value.
 - The time here is not discretized but the evolution of the system is broken up according to events.

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Modeling Space and Time

Time Dimension

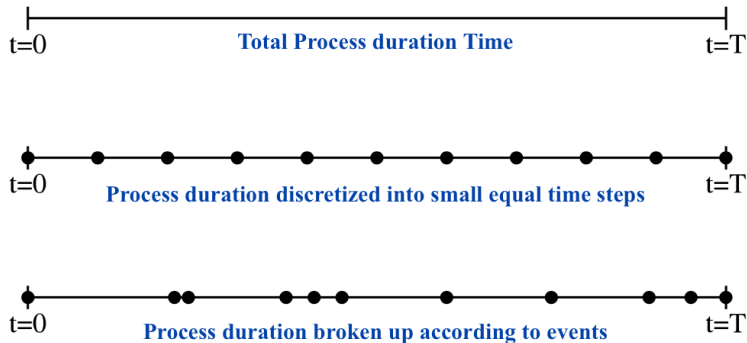


Figure 4: Temporal Dimension in a Model.

Introduction and General Concepts

Modeling Space and Time

Space Dimension

Space dimension can be included in a model using **Eulerian** or **Lagrangian** approach:

- In the **Eulerian** method, you take the point of view of an observer who sits at a fixed position $\vec{x}$ in space and just observes what happens at this specific position over time.
- e.g.: Recording the atmospheric pressure at this specific position at given time
- e.g.: counting every minute the number of cars that have crossed some line on the road in a traffic example.
- The **Eulerian** approach attach a physical property or state at each spatial locations.
- Space also supposed to be continuous (only mathematical models: PDE and ODE can describe that in a proper way).
- Otherwise, in computer model, we have to discretize the space into cells that usually makes a mesh, which covers the space in interest.

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Modeling Space and Time

Space Dimension

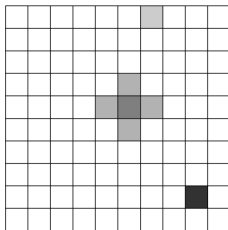
Space dimension can be included in a model using **Eulerian** or **Lagrangian** approach:

- In the **Lagrangian** method, you take the point of view of the objects moving in this space. Basically, the observer sits on one of these object and moves with it.
- e.g.: the movement of the Moon is described by its trajectory $\vec{x}(t)$, where $\vec{x}$ is a continuous variable.
- e.g.: in a traffic model, you can give the position of all cars over time.
- The **Lagrangian** approach take the point of view of the moving objets.

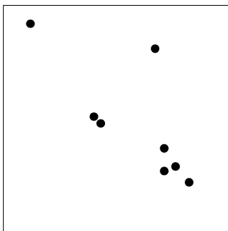
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Modeling Space and Time

Space Dimension



Eulerian point of view



Lagrangian point of view

Eulerian Method:

- The space (a square) here is divided in little cells. So, it is discretized in a mesh.
- A color describing the state of each position is given for each cell.
- e.g.: modeling the temperature - A White $\implies$ Cold, Black $\implies$ Warm, and Gray $\implies$ Between.

Lagrangian Method:

- Particles, for instance, particle can be anywhere in the space and you can describe them as their coordinates.

Introduction and General Concepts

Common Simulation Modeling Techniques

- Mathematical Equations: Ordinary Differential Equation (ODE), and Partial Differential Equation (PDE).
- Monte-Carlo method.
- Discrete Event Simulation.
- Multi-Agent Modeling Technique.
- Cellular Automata.
- Complex network

Introduction and General Concepts

Common Simulation Modeling Techniques

Mathematical Equations (ODEs and PDEs)

- Differential Equations (**DEs**) - Powerful mathematical tools to understand and predict the behavior of dynamical systems in nature, engineering, and society.
- **ODEs/PDEs** - A type of DE equations that specify the relations among the rate change (derivatives) of variables.

General Form of an ODE

$$0 = f\left(t, y, \frac{dy}{dt}, \frac{d^2y}{dt^2}, \dots, \frac{d^ny}{dt^n}\right) \quad (1)$$

- e.g.: 1st ODE would:

$$0 = f\left(t, y, \frac{dy}{dt}\right) \quad (2)$$

General Form of a PDE

$$0 = f\left(t, y, \frac{\partial y}{\partial t}, \frac{\partial^2 y}{\partial t^2}, \dots, \frac{\partial^n y}{\partial t^n}\right) \quad (3)$$

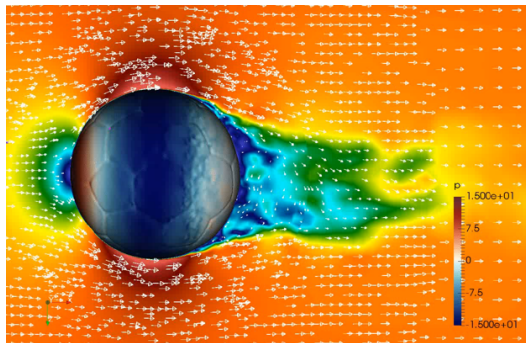
- e.g.: 1st PDE would:

$$0 = f\left(t, y, \frac{\partial y}{\partial t}\right) \quad (4)$$

Introduction and General Concepts

Common Simulation Modeling Techniques

Mathematical Equations (ODEs and PDEs)- e.g.: Fifa football Aerodynamics



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Common Simulation Modeling Techniques

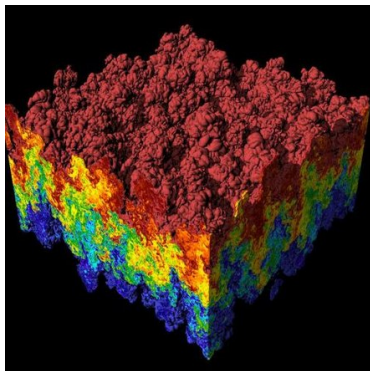
Monte-Carlo methods

- Monte Carlo methods - A broad class of computational algorithms that rely on repeated random sampling to obtain numerical results.
- Main idea - the results are computed based on repeated random sampling and statistical analysis.
- Used often in physical and mathematical problems and are most useful when it is difficult or impossible to use other approaches.

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Common Simulation Modeling Techniques

Monte-Carlo methods - e.g.: Large simulation of instability between two fluids of different densities.



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Common Simulation Modeling Techniques

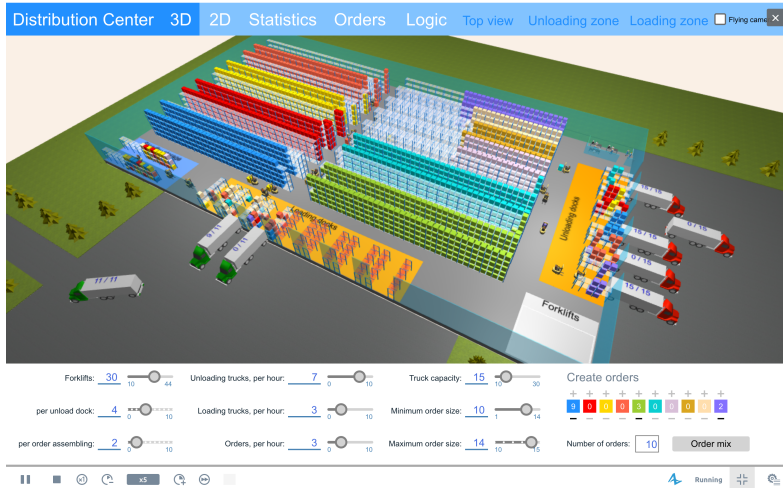
Discrete Event Simulation Techniques (DES)

- DES - A technique to models the operation of a system as a (discrete) sequence of events in time.
- Each event occurs at a particular instant in time and marks a change of state in the system.
- Between consecutive events, no change in the system is assumed to occur; thus the simulation time can directly jump to the occurrence time of the next event.

Introduction and General Concepts

Common Simulation Modeling Techniques

Discrete Event Simulation Techniques (DES) - e.g.: Distribution Centers



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Common Simulation Modeling Techniques

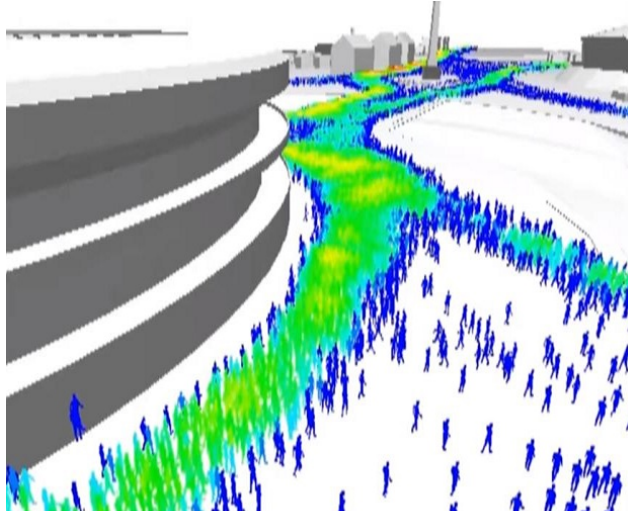
Multi-Agent Modeling Technique

- A style of modeling in which the interaction between (entities) individuals and with each other environment is modeled in a program.
- Main idea - Modeling the basic entities as individuals and observe the global emergent behavior.
- Agents can be, for example, people, animals, groups, or cells.

Introduction and General Concepts

Common Simulation Modeling Techniques

Multi-Agent Modeling Technique - e.g.: Pedestrian Crowd Simulation



Introduction and General Concepts

Common Simulation Modeling Techniques

Cellular Automata

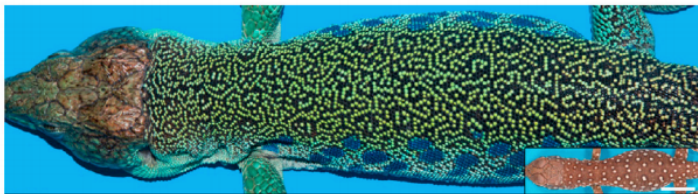
- A cellular automaton (CA) is a discrete model of computation technique.
- A CA consists of a regular grid of cells, each one is of a finite number of states, such as on and off.
- The grid can be in any finite number of dimensions (e.g.: 1D, 2D, 3D).
- For each cell, a set of cells called its neighborhood is defined relative to the specified cell.
- The grid is updated according to some fixed rule (generally, a mathematical function) that determines the new state of each cell.
- Can be applied in Generative art and music, Skin Pattern formation, Biology process, ...

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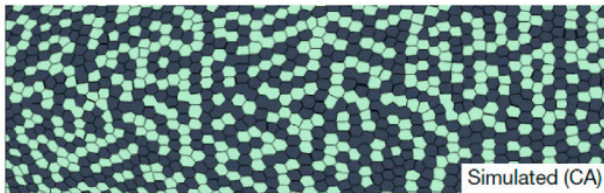
Common Simulation Modeling Techniques

Cellular Automata - e.g.: A Model for Skin Color Patterning in Lizards

A



B



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Common Simulation Modeling Techniques

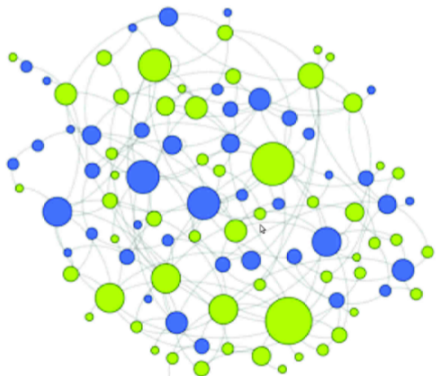
Complex Network

- A complex network is a mathematical representation of measurable variables and its interactions.
- The right way to represent this in a model is to use a Graph, which can be also evolving in time (Dynamic).
- Used essentially to see whether patterns may emerge from the interactions between the nodes.
- Can be applied to a variety of studies involving inflammatory interactions in a cell, human diseases, social networking, ...

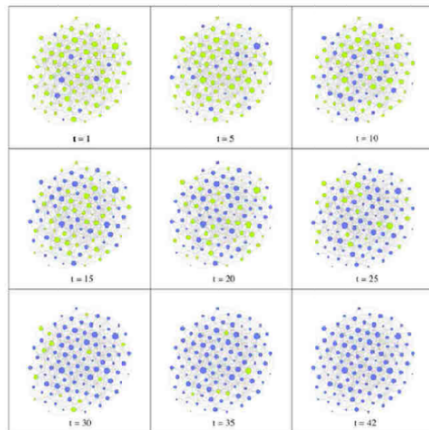
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Common Simulation Modeling Techniques

Complex Network - e.g. of a Model on Opinion Propagation in a Social Network



(Lino Velasquez, UNIGE)



Réseau aléatoire, $N = 100$, $p = 0.05$, $\epsilon = 0.3$

End of Lecture 1:

Introduction and General Concepts